DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION

✈ Chapter 8: VOR

VHF Omnidirectional Range

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1. Introduction & Background

VOR — **VHF Omnidirectional Range** — is the primary short-range radio navigation aid used worldwide. It provides magnetic bearing information to/from a ground station and forms the backbone of conventional airway navigation.

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Figure 8.1 A combined VOR/DME

The VHF Omni-directional Range (VOR) was adopted as the standard short range navigation aid in 1960 by ICAO. It produces bearing information usually aligned with magnetic north at the VOR location. It is practically free from static interference and is not affected by sky waves, which enables it to be used day and night. When the VOR frequency is paired with a co-located Distance Measuring Equipment (DME) an instantaneous range and bearing (Rho-Theta) fix is obtained. The equipment operates within the frequency range of 108 - 117.95 MHz.

VOR has the following uses:

- Marking the beginning, the end and centre line of airways or sections of airways.
- As a let-down aid at airfields using published procedures.
- As a holding point for aircraft.
- As a source of en route navigational position lines.

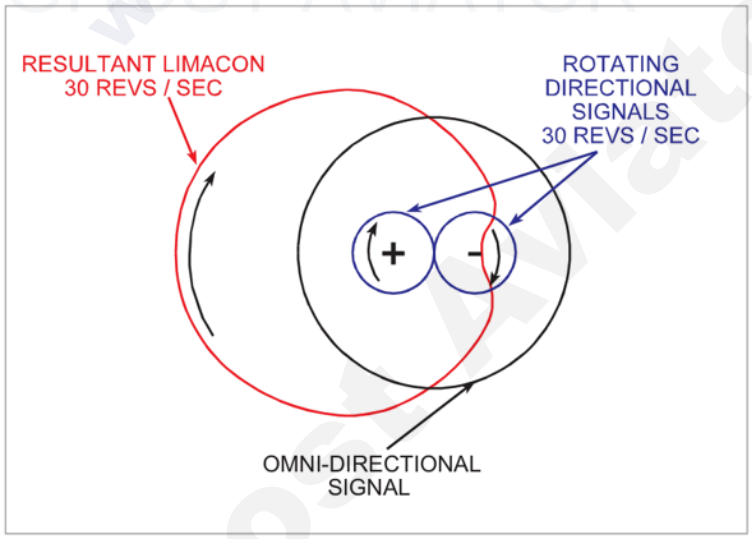


Figure 8.2 A VOR Polar Diagram

Fig 8-1: VOR/DME station with omnidirectional polar diagram

Key Facts

- Adopted by
  ICAO in 1960
  as the standard short-range navaid
- Operates in the
  VHF band
  — immune to static interference and skywave
- Provides
  magnetic bearing
  to/from the station (azimuth service)

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- Emission designator: **A9W**
- Co-located with DME → **VOR/DME**; with TACAN → **VORTAC**

FREQUENCY BAND	TOTAL CHANNELS
108.0 – 117.95 MHz	**160 channels**

SHARED WITH ILS	VOR ONLY
108.0 – 111.95 MHz (40 ch)	**112.0 – 117.95 MHz (120 ch)**

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2. Principle of Operation

The VOR works by **phase comparison** of two 30 Hz signals. The phase difference between the two signals equals the aircraft's magnetic bearing from the station (radial).

🔄 The Two Signals

Signal	Name	Modulation	Characteristic
Reference	Omnidirectional reference	FM subcarrier (9960 Hz)	Same phase in ALL directions
Variable	Directional / variphase	AM on carrier	Phase varies with azimuth (limàcon pattern)

The variable signal is created by rotating a limàcon (figure-of-eight) radiation pattern at **30 rev/sec**. At **magnetic north**, both signals are in phase. Moving clockwise, the variable signal phase progressively leads the reference — the phase difference equals the radial.

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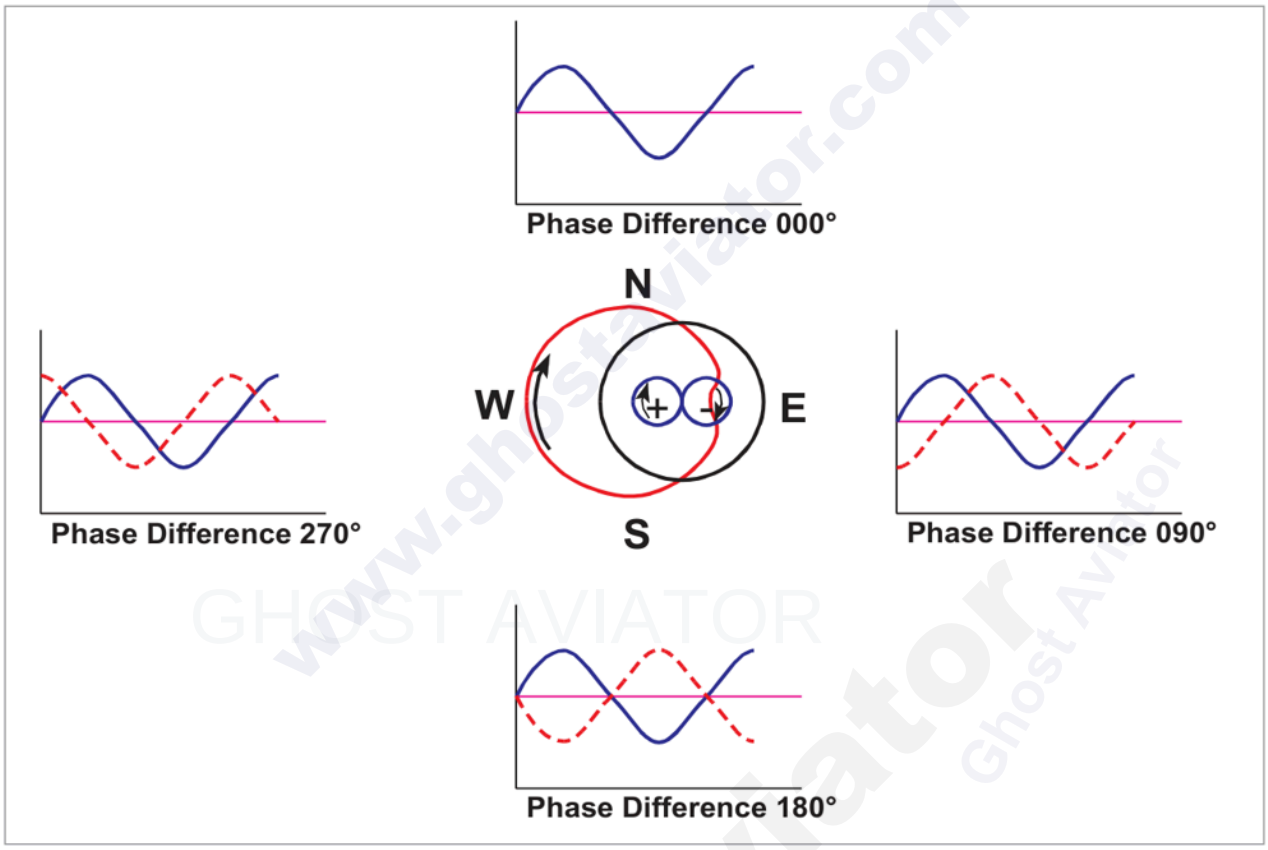


Figure 8.3 Phase differences corresponding to the cardinal points

Fig 8-3: Phase differences at cardinal points (N=0°, E=90°, S=180°, W=270°)

> ⚡ **Exam Tip — Phase = Radial**
>
> Phase difference (variable − reference) = Radial (QDR) = Magnetic bearing FROM the VOR.
>
> Example: Aircraft due East → phase difference = **90°**

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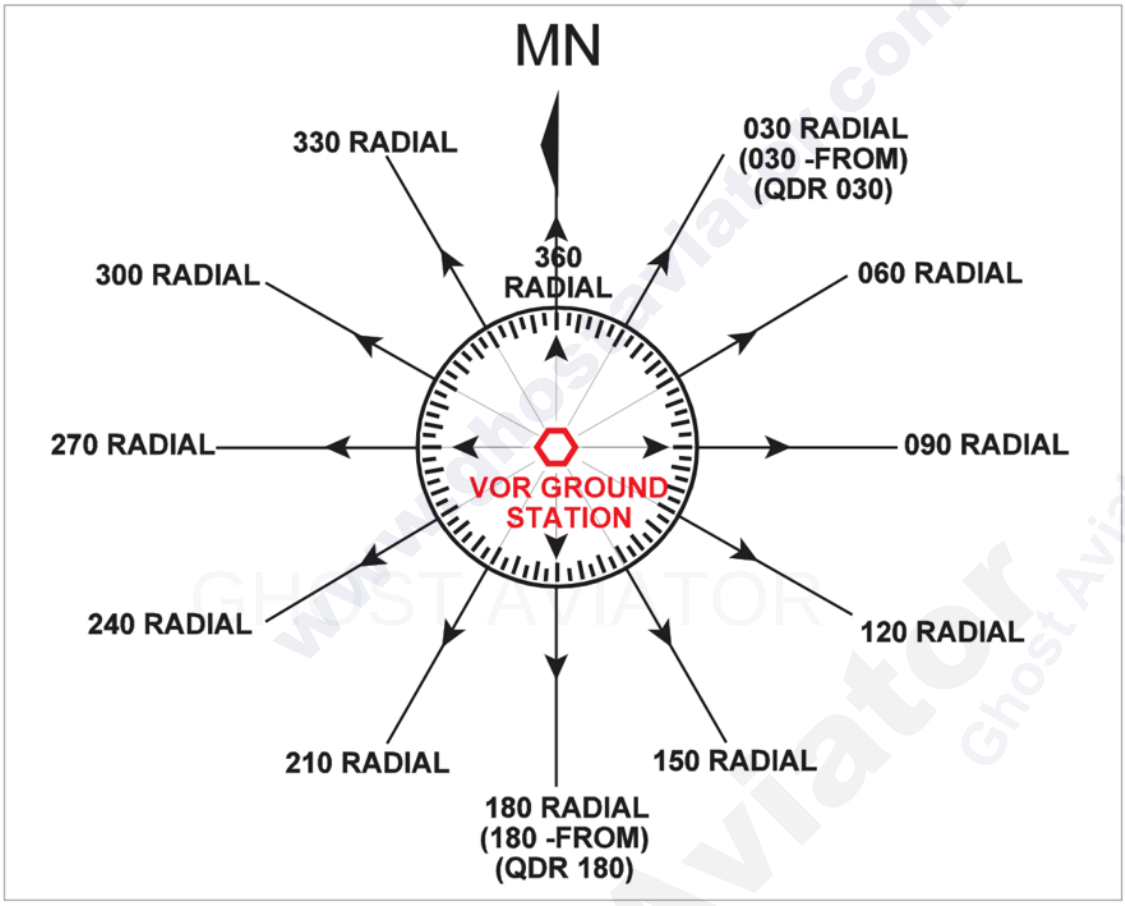


Figure 8.4 A Radial is a Magnetic Bearing from the VOR (i.e. QDR)

Fig 8-4: Radial definition — phase difference equals magnetic bearing FROM station

CVOR vs DVOR Signal Generation

Feature	CVOR (Conventional)	DVOR (Doppler)
Reference signal	FM subcarrier (9960 Hz ±480 Hz)	AM on carrier
Variable signal	AM on carrier (rotating limàcon CW)	FM via Doppler (simulated ACW rotation)
Site error	More susceptible	Less site error — preferred at complex sites
Antenna array	Single rotating antenna	Ring of 48 antennas, switched electronically

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3. Terminology — Radials & Bearings

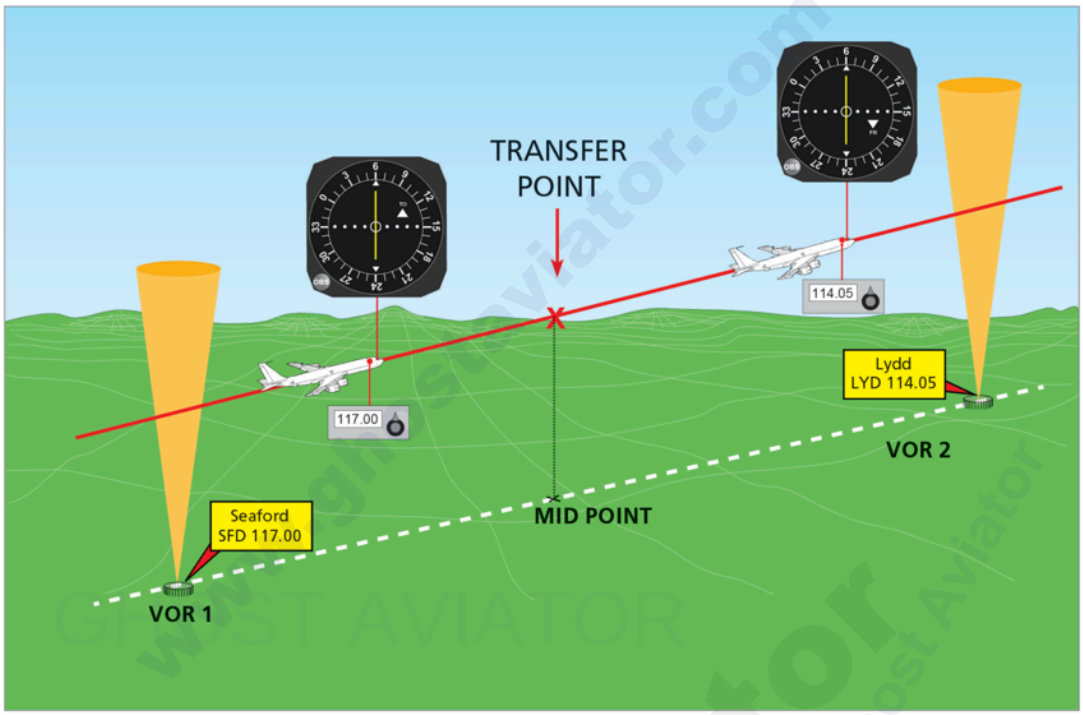


Figure 8.5 Tracking Between Two VORs



Figure 8.6 RMI Usage

Fig 8-5: Tracking between two VORs and RMI usage

Critical Definitions

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Term	Definition	Note
Radial	Magnetic bearing FROM the VOR (QDR)	What the VOR transmits
QDR	Magnetic bearing FROM station to aircraft	RMI *tail*
QDM	Magnetic heading to fly TO station (no wind)	RMI *head*
QTE	True bearing FROM station	Apply variation at VOR

⚡ Variation Rule

For VOR bearings, apply variation at the
VOR station
to convert magnetic to true bearing.
This is opposite to ADF/NDB where variation is applied at the aircraft.

✓ Example

Aircraft on radial 090° from a VOR where local variation = 10°W
QDR = 090°M → QTE = 090° − 10° = **080°T** (subtract westerly variation)

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4. Frequencies & Channels

Band	Range	Channels	Notes
Lower VOR band	108.000 – 111.975 MHz	40 channels	Shared with ILS Localiser — **even tenths** = VOR (108.0, 108.2…); odd tenths = ILS
Upper VOR band	112.000 – 117.975 MHz	120 channels	VOR exclusive

⚡ Frequency Sharing (108–112 MHz)

Even tenths

(108.0, 108.2, 108.4…) = VOR |

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Odd tenths

(108.1, 108.3...) = ILS Localiser.

This allows 40 VOR + 40 ILS channels in the shared band.

Emission Designator: A9W

A = Amplitude modulation (DSB full carrier) | 9 = Composite signal | W = Combination (telephony + telegraphy + navigation)

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5. Identification & Monitoring

Station Identification

Ident Methods

- 3-letter Morse code transmitted every **10 seconds** on a 1020 Hz subcarrier
- Some stations also broadcast voice ATIS on the VOR frequency
- Keyed at **7 words per minute**

Special Beacon Types

Type	Ident Signal	Purpose	Use?
TST	TST in Morse	Test beacon — under maintenance/calibration	✗ **DO NOT USE**
VOT	Continuous dots	VOR Test facility — checks aircraft VOR equipment	✓ **Ground test only**

⚠ VOT Test Limits

When checking with a VOT, the aircraft must read **000° ± 4°** (FROM) or **180° ± 4°** (TO).

Maximum permissible error = **±4°**

Monitoring

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🔍 Automatic Site Monitor

- Built-in monitor checks VOR bearing accuracy continuously
- If bearing error exceeds **±1°** → station shuts down or switches to standby transmitter
- During standby switchover:
  ident is suppressed for 30 seconds
- Rule:
  NEVER USE a VOR when no ident is heard

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6. Types of VOR

Type	Full Name	Power	Use
CVOR	Conventional VOR	Standard	En-route navigation
DVOR	Doppler VOR	Standard	Sites with terrain/structure interference
TVOR	Terminal VOR	Low (≤50W)	Airfields — short range approach/departure
VOT	VOR Test facility	Very low	Ground equipment check only
VORTAC	VOR + TACAN	Standard	Civil (VOR) + Military (TACAN bearing + DME)
VOR/DME	VOR + DME	Standard	Provides bearing + distance

📡 DVOR — Doppler Mechanism

- AM carrier =
  reference signal
- Ring of
  48 antennas
  switched electronically to simulate a rotating antenna
- Doppler effect creates the
  FM variable signal
- Simulated rotation is
  anticlockwise

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(opposite to CVOR) — receiver processes identically

- Less susceptible to multipath site errors than CVOR

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7. Range & Coverage

VOR is a **line-of-sight (LOS)** system. Range depends on aircraft altitude and terrain.

LOS Range Formula

$$\text{Range (NM)} = 1.25 \times (\sqrt{H_{tx}} + \sqrt{H_{rx}})$$

Where H = height in

feet

above terrain.

Note: Coefficient **1.25** is used for all DGCA exam calculations (some texts show 1.23, but exam answers verify with 1.25).

✓ Example

Aircraft at 10,000 ft, VOR antenna at 100 ft above terrain:

Range = 1.25 × (√10000 + √100) = 1.25 × (100 + 10) = 1.25 × 110 = **137.5 NM**

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8. DOC — Designated Operational Coverage

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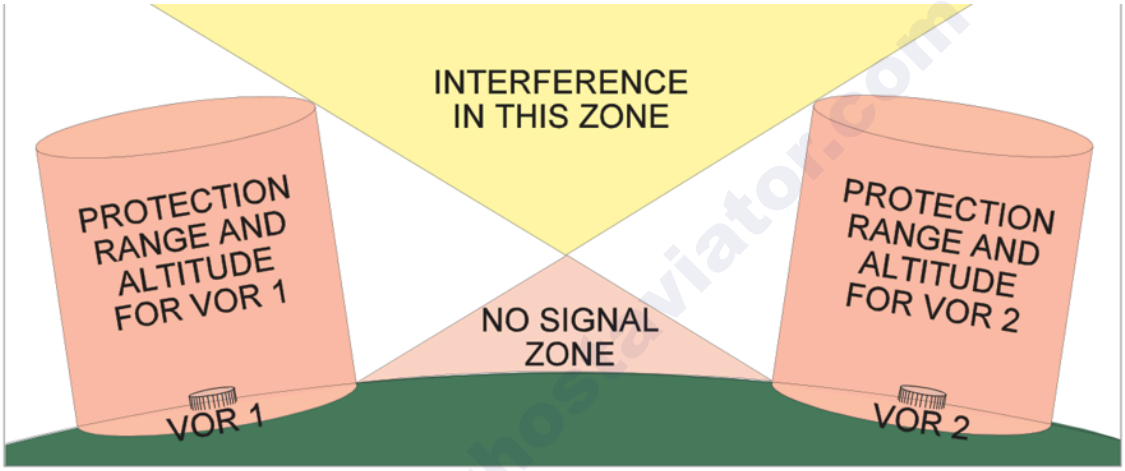


Figure 8.7 Designated Operational Coverage.

VOR 1: DOC 50/25 = No interference within 50 NM range up to 25 000 ft.

VOR 2: DOC 100/50 = No interference within 100 NM range up to 50 000 ft.

Fig 8-7: DOC diagram showing range/altitude pairs

📐 DOC Definition

- Specifies the
  range and altitude
  within which the VOR provides reliable navigation service
- Published as a range/altitude pair — e.g., **50 NM / FL 250**
- Means: reliable up to 50 NM from the station at or below FL 250
- Valid
  day AND night
  (unlike NDB/ADF which has a day-only DOC)

✓ VOR DOC vs ADF DOC

VOR DOC is valid day and night — VHF is not affected by skywave.

NDB/ADF DOC is
day only
— MF/LF skywave at night causes bearing errors.

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9. Accuracy & Errors

Error Source	Value	Notes
Site/terrain error	± **1°**	Multipath reflections; DVOR less susceptible
Propagation / scalloping	± **1°**	Bending of VHF waves; needle oscillation in affected areas
Airborne equipment	± **3°**	Receiver and aircraft installation accuracy
Total system error	**± 5°**	ICAO specification (RSS combination)

⚠ Airway Width — 1:60 Rule

At distance D, lateral error = D × error° / 60.

To maintain ±5 NM half-width with ±5° error: D = (5 × 60) / 5 = **60 NM** per leg?

Standard formula for max beacon spacing: max_spacing = 600 / error_degrees

• For ±5° error: 600/5 = **120 NM** • For ±5.5° error: 600/5.5 ≈ **109 NM**

⚡ Scalloping

Rapid oscillation of VOR bearing indication caused by multipath interference. The CDI needle "scallops" back and forth. More pronounced in mountainous terrain; DVOR is less susceptible.

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10. Cone of Confusion

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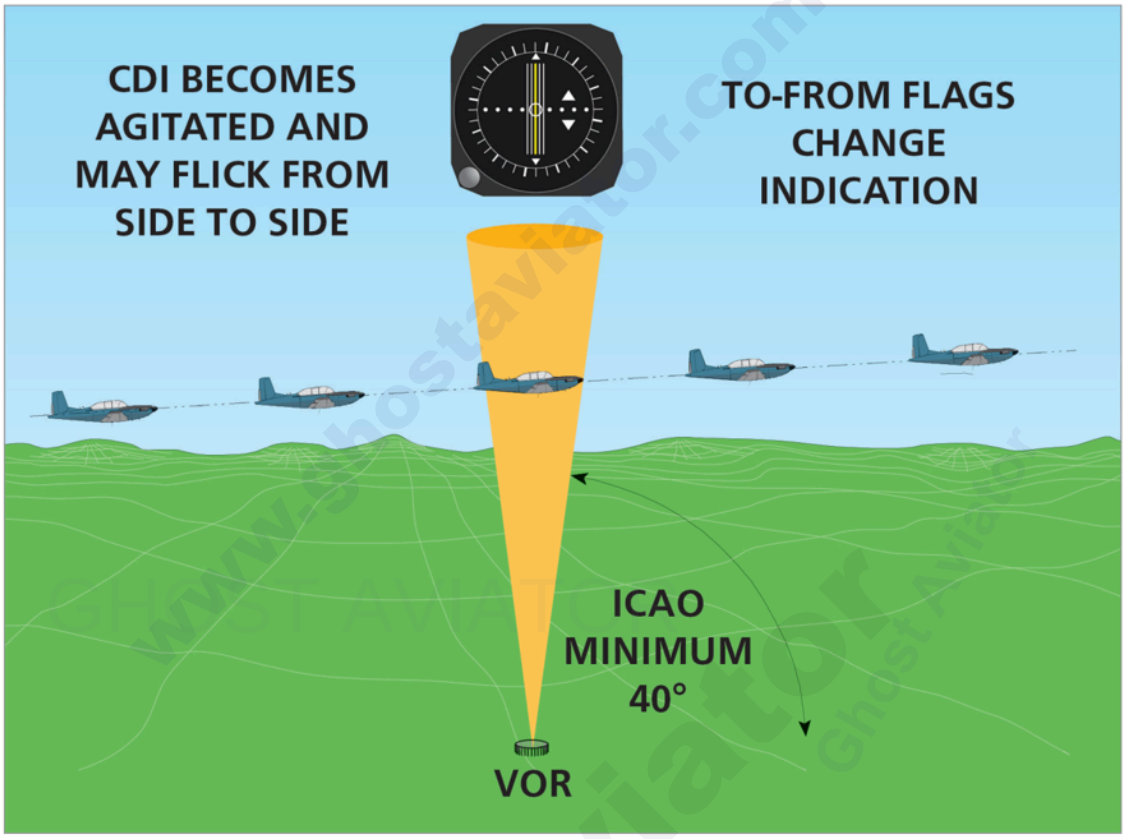


Figure 8.8 The Cone of Confusion

Fig 8-8: Cone of confusion — unreliable signal area directly overhead the VOR

⚠ Cone of Confusion — Overhead VOR Indications

- OFF flag may appear on CDI/VOR indicator
- TO/FROM indicator alternates rapidly
- CDI needle oscillates / swings
- Extends approximately ±50° from vertical (antenna design dependent)
- Normal and temporary — do not attempt corrections while in the cone

✓ Station Passage Technique

1. Note the moment TO/FROM flag flips to FROM → that is station passage.
2. CDI will settle after exiting the cone (typically within 1–2 minutes at normal speeds).

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11. Cockpit Displays — CDI, RMI, HSI

11.1 CDI — Course Deviation Indicator



Figure 8.9 VOR/ILS Deviation Indicator

Fig 8-9: CDI / OBS (Omni Bearing Selector) deviation indicator

bar would be central.

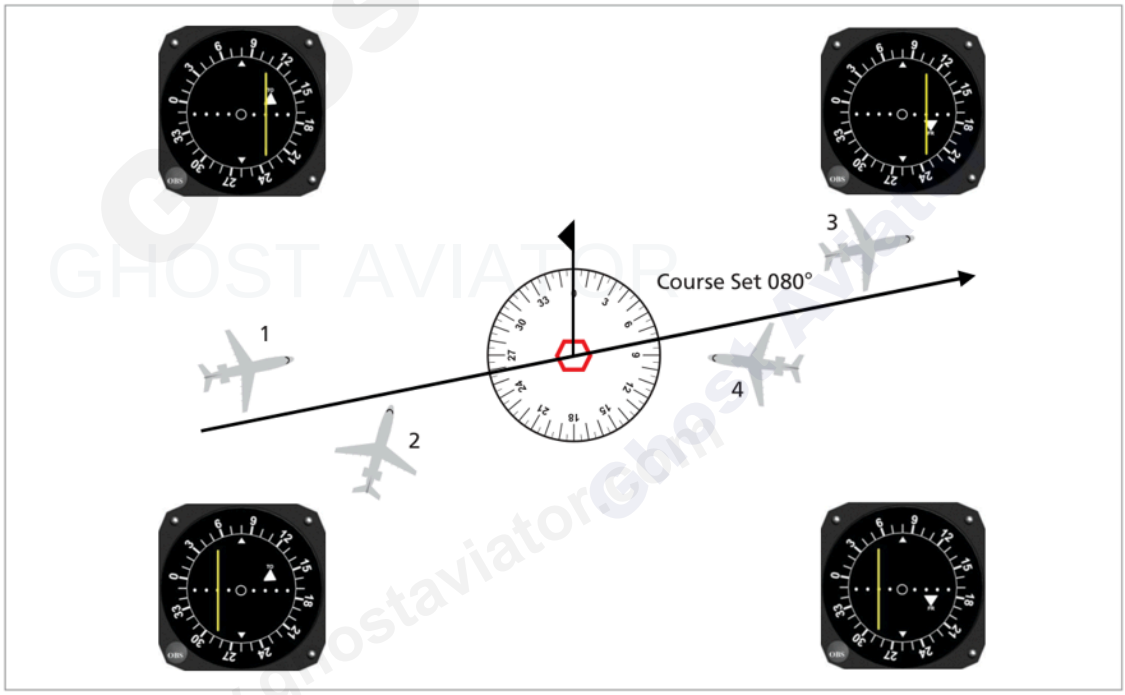


Figure 8.10 Left/Right Indications

Fig 8-10: Left/Right CDI indications for various radial/OBS combinations

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obtained.

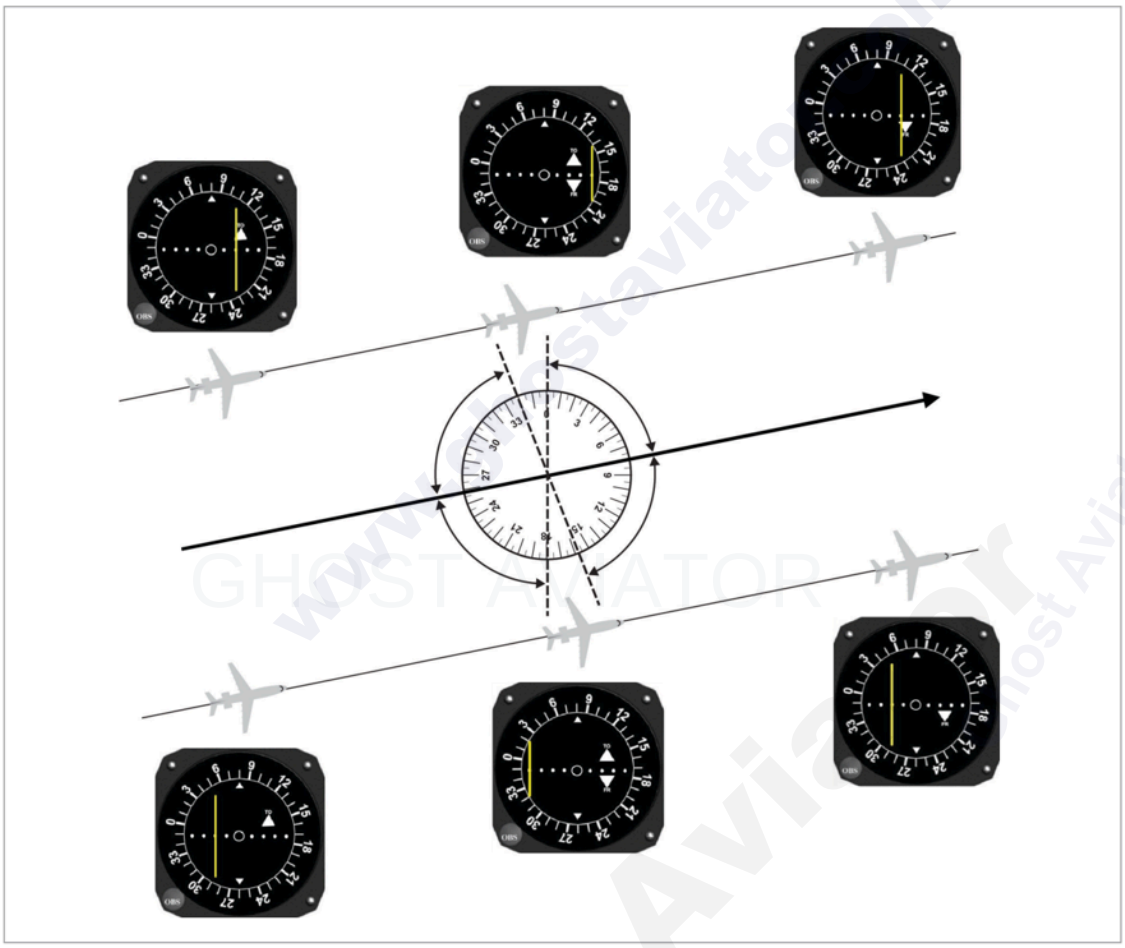


Figure 8.11 To/From Indications

Fig 8-11: TO/FROM indications on CDI

📐 CDI Key Facts

Parameter	Value
Sensitivity	**2° per dot**
Full Scale Deflection (FSD)	**±10°** (= ±5 dots)
TO indication	QDM is within ±80° of OBS setting
FROM indication	QDR is within ±80° of OBS setting
OFF indication	OBS is ~90° away from QDM/QDR (ambiguous zone)

⚡ CDI Sense — Critical Rule

The CDI only shows correct turn direction when the

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OBS setting matches your actual desired track

. If OBS is set wrong, the needle sense can be reversed — this is the classic DGCA CDI trap!

⚠ CDI Worked Example

Aircraft on 152° radial, OBS = 329° (TO). Desired radial = 149°. Aircraft is 3° to the RIGHT of desired track.

Therefore aircraft is to the LEFT of the 329° course → CDI bar deflects **RIGHT** (fly right).

Deflection = 3° ÷ 2°/dot = **1.5 dots right**.

11.2 RMI — Radio Magnetic Indicator

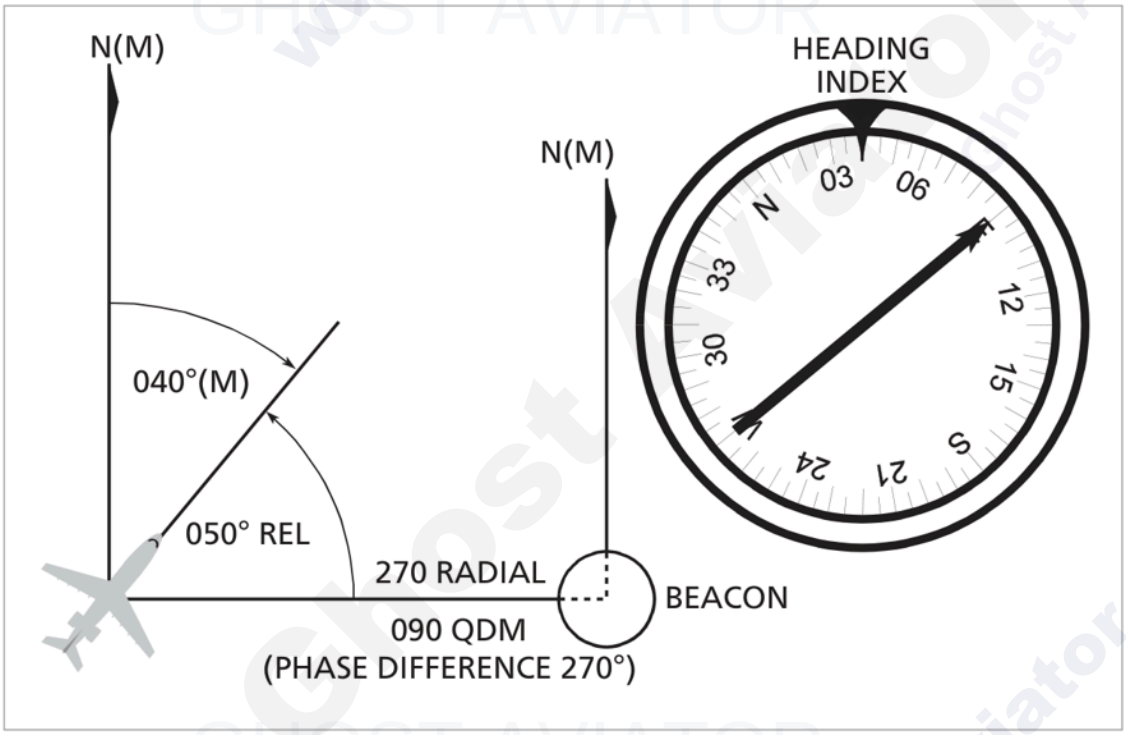


Figure 8.12 VOR QDM on the RMI

Fig 8-12: RMI reading example — phase difference to bearing conversion

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RMI (Radio Magnetic Indicator)



Figure 8.13

Heading displayed will always be Compass Heading, which should be very close (deviation) to Magnetic Heading.

The VOR QDR/QDM displayed will be the Radial that the aircraft is actually on. *(For examination purposes; If the arrowhead and the tail do not agree, due to a bent needle, then the arrowhead will be the correct reading)*

The arrowhead will always point TO the beacon QDM.

The aircraft's heading will not affect the readings.

If heading is correct then both Relative Bearing and Radial will be correct.

If heading is in error then the Radial will be correct but the Relative Bearing will be wrong.

No action is required by the pilot.

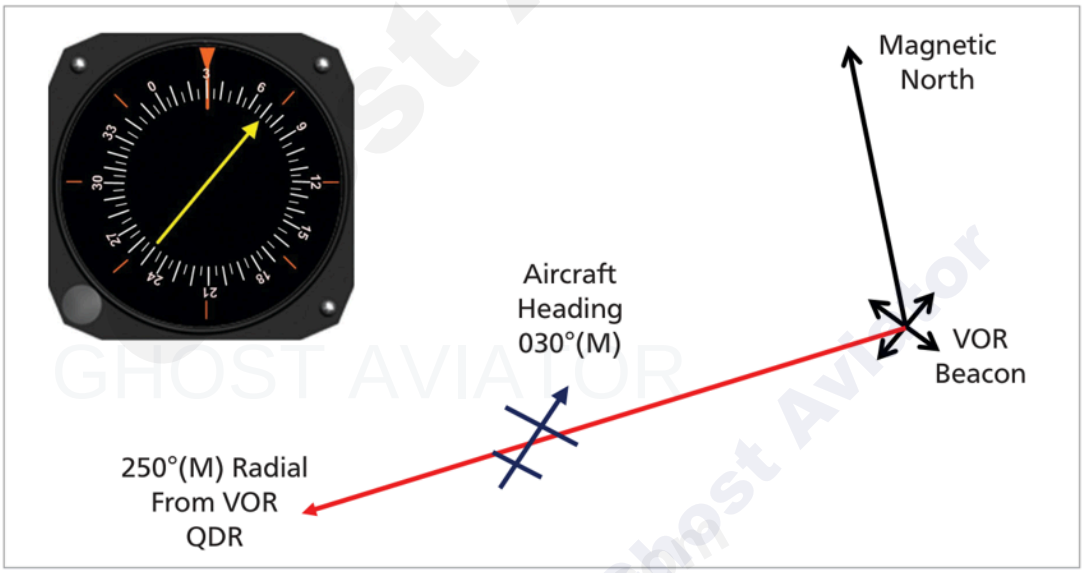


Figure 8.14

Fig 8-13: RMI display and geometry

RMI Key Facts

Needle Part	Reading	Equivalent

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Head (arrowhead)	Points TO the station	QDM — magnetic bearing TO station
Tail	Points AWAY from station	QDR — radial — magnetic bearing FROM station

The RMI compass card rotates with aircraft heading, so the needle always reads **magnetic bearings**

(not relative bearings).

✓ RMI Worked Examples (from page 128 Q1–4)

1. Using the CDI shown, what is the aircraft's QDR?



Figure 8.19

2. Using the CDI shown, what is the aircraft's QDM?



Figure 8.20

3. Using the HSI shown, what is the aircraft's QDR?



Figure 8.21

4. Using the HSI shown, what is the aircraft's QDM?



Figure 8.22

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Q1-4: Four RMI diagrams for worked examples

- Q1: Heading 360°, needle head at 071° → QDM = **071°**, QDR = 251°
- Q2: Heading 090°, needle head at 159° → QDM = **159°**, QDR = 339°
- Q3: Heading 220°, needle head at 345° → QDM = **345°**, QDR = 165°
- Q4: Heading 270°, needle head at 063° → QDM = **063°**, QDR = 243°

11.3 HSI — Horizontal Situation Indicator

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Figure 8.15

Course shown at top/centre of the dial is the Required Course to fly to achieve the desired aim.

The TO/FROM indicator will be decided by the instrument. If the actual radial, which the aircraft is on, is within 90° of the Set Course then FROM will be shown. If the actual radial, which the aircraft is on, is more than 90° from the Set Course then TO will be shown.

The Course Deviation Bar shows the angular difference between the Required Course and the actual VOR Radial the aircraft is on.

1 Dot = 2° **Full Scale Deflection = 10°**

The edge of the inner circle is the first dot = 2°

The aircraft's heading will not affect the readings.

The pilot sets the Required Course, using the OBS knob.

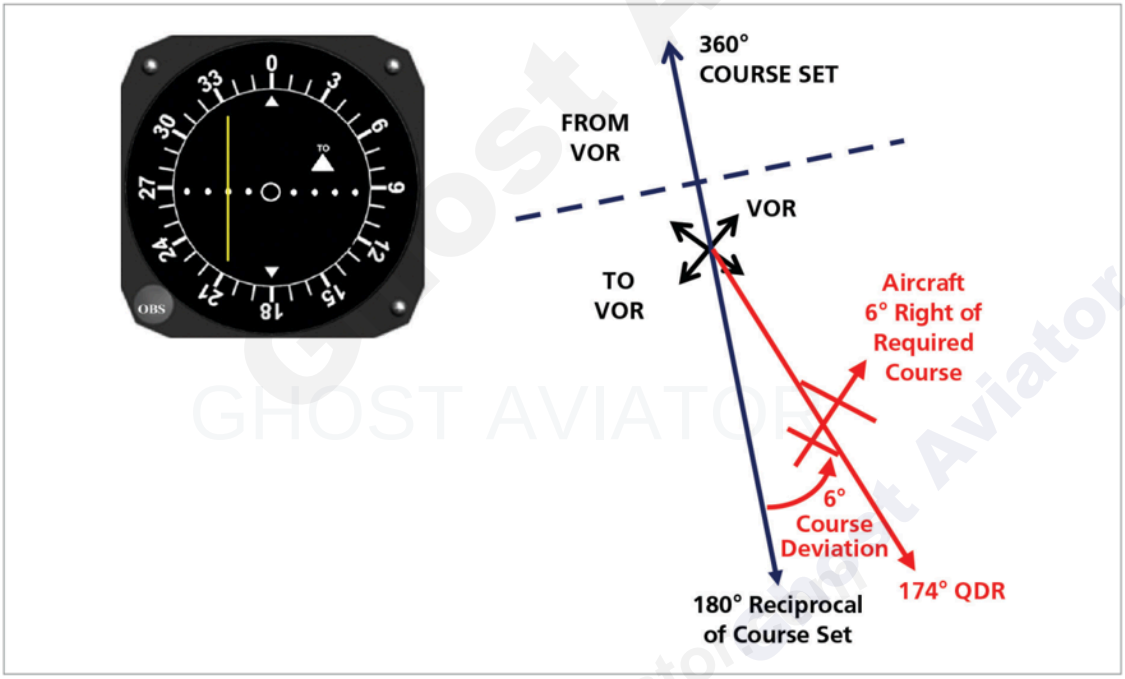


Figure 8.16

Fig 8-15: CDI deviation bar within HSI display

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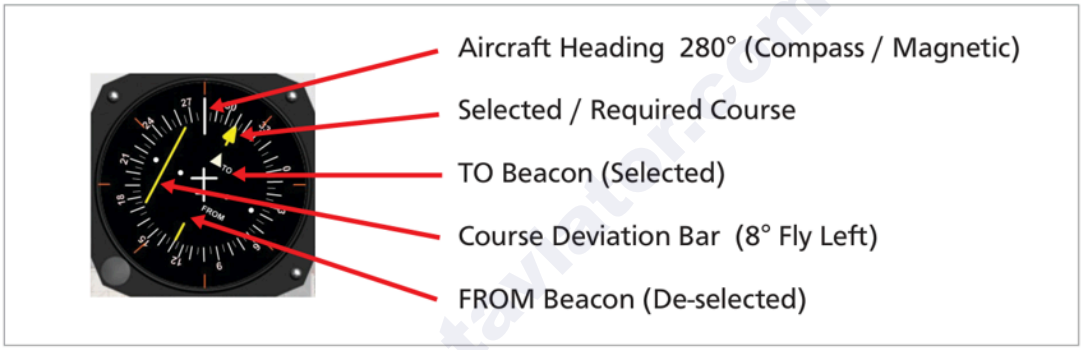


Figure 8.17

Heading displayed will always be Compass Heading, which should be very close (deviation) to Magnetic Heading.

The arrowhead shows the Required Course, set by the pilot.

The TO/FROM indicator will be decided by the instrument. If the actual radial, which the aircraft is on, is within 90° of the Set Course then FROM will be shown. If the actual radial is more than 90° from the Set Course then TO will be shown.

The Course Deviation Bar shows the angular difference between the Required Course and the actual VOR Radial the aircraft is on.

1 Dot = 5° **Full Scale Deflection = 10°**

An HSI may be either a 2 Dot or 5 Dot display
Full Scale Deflection will always be 10°. 2 Dot: display 1 dot = 2°. 5 Dot display: 1 Dot = 5°

Aircraft heading is taken into consideration in displaying a fly left or fly right indication. However, as the instruments includes heading, it is able to determine the best direction to turn to achieve the required radial. So it is possible to be right of the radial but to be given a turn right indication.

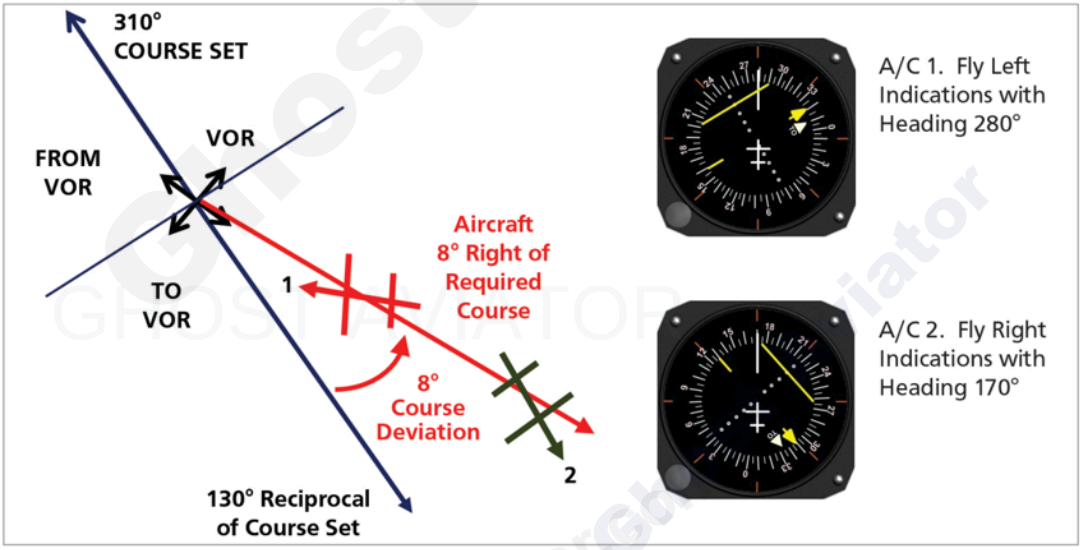


Figure 8.18

Fig 8-17: Full HSI display with course pointer and heading

HSI Key Facts

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Parameter	Standard HSI	Narrow-bar HSI
Sensitivity	**5° per dot**	**2° per dot**
Dots each side	2 dots	2 dots
FSD	±10°	±4°

The HSI course pointer rotates with aircraft heading → the deviation bar **always shows correct turn direction** regardless of OBS setting. No CDI sense reversal.

◆ CDI vs RMI vs HSI Comparison

Summary

RMI



Aircraft Heading 030° (Compass / Magnetic)
QDM – Arrowhead 070°
QDR – Tail of Needle 250°
Aircraft's Heading is NOT Relevant

CDI



Selected / Required Course
TO Beacon (Selected)
FROM Beacon (De-selected)
Course Deviation Bar (6° Fly Left)
Aircraft's Heading is NOT Relevant

HSI



Aircraft Heading 280° (Compass / Magnetic)
Selected / Required Course
TO Beacon (Selected)
Course Deviation Bar (8° Fly Left)
FROM Beacon (De-selected)
Aircraft's heading will determine Turn Left / Turn Right Indication

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Summary comparison: RMI, CDI, and HSI displays side by side

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12. In-Flight Procedures

Typical uses of VOR by an aircraft equipped with both CDI-type deviation indicator and an RMI are illustrated in *Figure 8.23*.

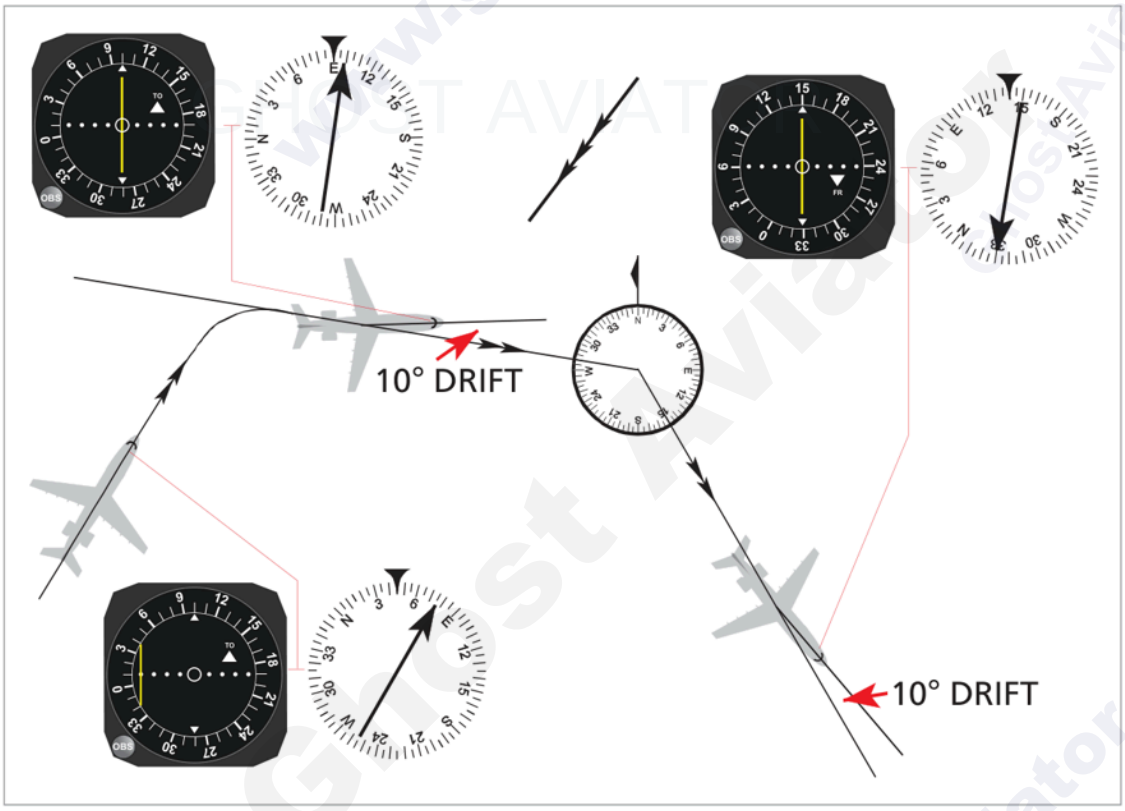


Figure 8.23 In-flight Procedures

Fig 8-23: In-flight VOR tracking procedures

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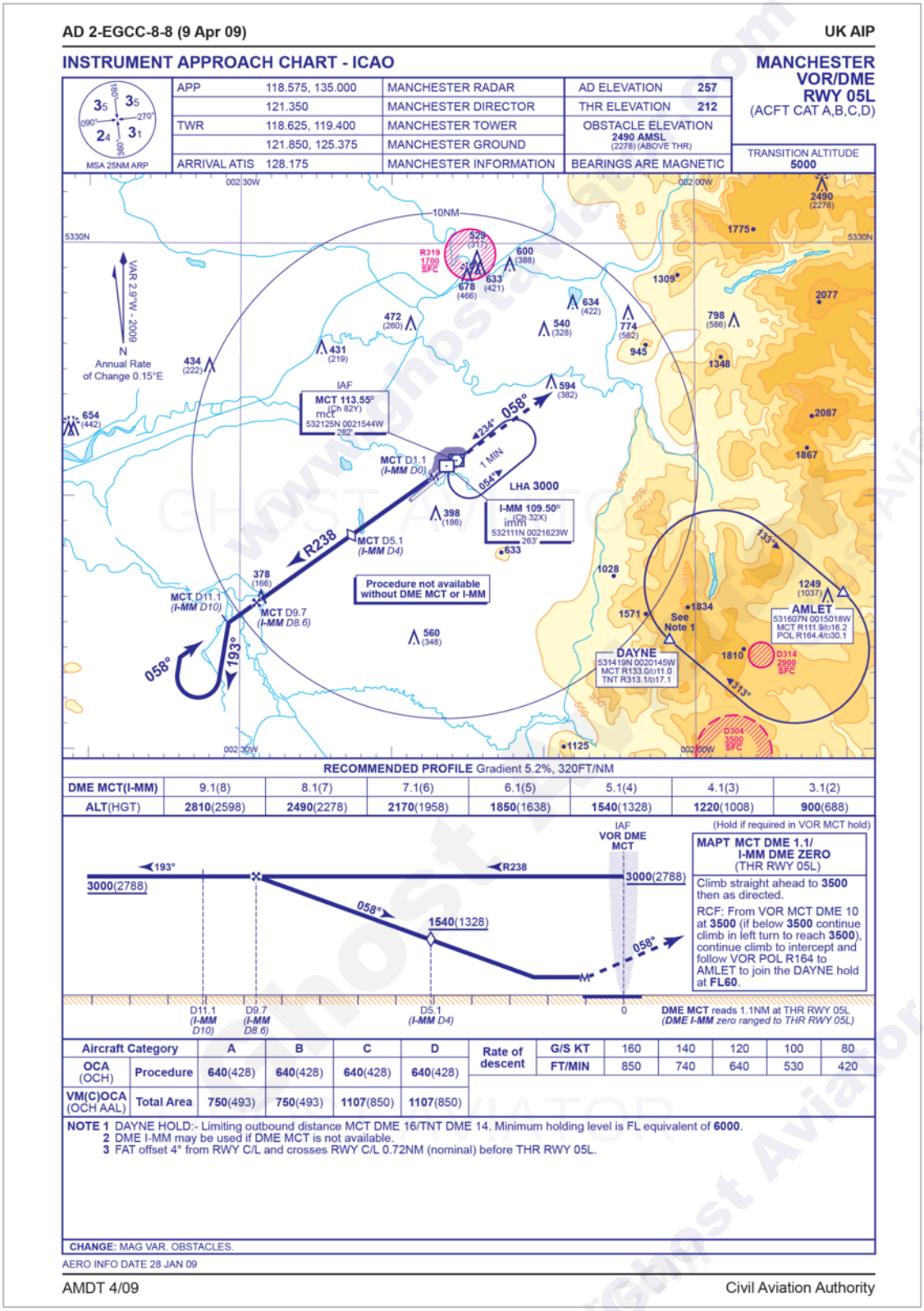


Figure 8.24 Example of a VOR DME approach pattern

Fig 8-24: VOR/DME approach procedure

✓ Outbound Tracking (away from VOR)

1. Set OBS to desired outbound radial (e.g., 090° for east outbound)

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2. FROM flag appears
3. Wind correction: CDI drifts right → turn right to correct (normal sense with FROM selected)

✓ Inbound Tracking (toward VOR)

1. Set OBS to inbound track (reciprocal of radial, e.g., 270° for inbound on 090° radial)
2. TO flag appears
3. CDI normal sense — fly toward the needle

📡 Station Passage Indications

- Primary: TO → FROM flag change
- CDI needle may swing briefly
- OFF flag may appear momentarily in cone of confusion
- Log time of station passage for flight planning

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13. VOR Summary Table

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VOR Summary

Characteristics:	Magnetic bearings, valid day and night
Frequency:	108 to 117.95 MHz; 160 channels
Uses:	Airways; Airfield let-downs; Holding points; En route navigation
Principle of Op:	Phase comparison of two 30 Hz signals
Identification:	3 letter aural Morse or Voice every 10 s, continuous tone for VOT (also ATIS using AM on voice)
Monitoring:	Automatic site monitor +/- 1° Ident suppressed when standby transmitter initially switched on
Types:	CVOR - reference signal is FM; variphase signal is AM - Limacon polar diagram rotating clockwise DVOR - more accurate than CVOR due to less site error - reference signal is AM; variphase signal is FM - simulated anticlockwise rotation of aerial TVOR - low power Tx at airfields VOT - Test VOR giving 180 radial - aircraft should have < +/- 4° error
Operational range:	Transmitter power Line of sight DOC valid day and night
Accuracy affected by:	Site error (less with DVOR) Propagation error Scalloping (bending due to reflections from terrain) Airborne equipment error (+/- 3°)
Cone of confusion:	OFF flag may appear; TO/FROM display and bearings fluctuate
Airborne equip:	Aerial, Receiver, Display (CDI/RMI) CDI: 2° per dot; max 10°; relationship between indication and aircraft position RMI: arrowhead gives QDM; tail gives QDR; Use magnetic variation at station
In-flight procedures:	Radial interceptions; Track-keeping; Station passage

Complete VOR Summary Table

Parameter	Value / Detail
Frequency band	108.0 – 117.95 MHz (VHF)
Total channels	160 (40 shared with ILS + 120 VOR only)
Emission designator	A9W
Principle	Phase comparison of two 30 Hz signals
Ident interval	3-letter Morse every 10 s (1020 Hz tone)
Monitor trigger	±1° bearing error → auto-shutdown
VOT tolerance	±4°

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Site error	±1°
Total system accuracy	±5°
DOC validity	Day AND night
LOS range formula	1.25 × ($\sqrt{H_1}$ + $\sqrt{H_2}$) NM (H in feet)
CDI sensitivity	2°/dot, FSD ±10°
HSI sensitivity	5°/dot standard (FSD ±10°); 2°/dot narrow
Radial (QDR)	Magnetic bearing FROM VOR to aircraft
Variation applied at	VOR station (not aircraft)
Cone of confusion	±50° from vertical — OFF flag, needle oscillates

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14. Practice Questions

⚡ **Instructions**

Click “Show Answer” to reveal the correct answer and explanation. All questions are DGCA CPL/ATPL standard.

Q1. What is the frequency band used by VOR?

(a) 225–400 MHz

(b) 108–117.95 MHz

(c) 190–1750 kHz

(d) 962–1213 MHz

Answer: (b)

VOR operates in the **VHF band, 108.0–117.95 MHz**. (a) = UHF military TACAN; (c) = NDB/ADF MF/LF band; (d) = DME UHF.

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Q2. The principle of operation of a VOR is:

(a) Comparison of two 9960 Hz signals

(b) Pulse timing like DME

(c) Phase comparison of two 30 Hz signals

(d) Amplitude comparison of two signals

> **Answer: (c)**
> VOR uses **phase comparison of two 30 Hz signals**. The reference is omnidirectional (FM subcarrier); the variable rotates (AM on carrier). Their phase difference = radial.

Q3. A VOR radial is defined as:

(a) Magnetic heading to fly to the station

(b) True bearing from aircraft to station

(c) Magnetic heading of the aircraft

(d) Magnetic bearing FROM the station to the aircraft (QDR)

> **Answer: (d)**
> A radial = **QDR = magnetic bearing FROM VOR to aircraft**. Aircraft on the 090° radial is due east of the station.

Q4. A VOR station antenna is at 100 ft; aircraft at 3,600 ft. Using coefficient 1.25, maximum LOS range is:

(a) 87.5 NM

(b) 75 NM

(c) 100 NM

(d) 62.5 NM

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Answer: (a)
Range = 1.25 × (√3600 + √100) = 1.25 × (60 + 10) = 1.25 × 70 = **87.5 NM**.

Q5. Which statement about a VOT is correct?

(a) It radiates a continuous tone and is used to check aircraft VOR equipment accuracy on the ground

(b) It is a type of VOR for terminal area use

(c) It stands for 'VOR on Test' and should not be used

(d) It provides bearing guidance to the test facility

Answer: (a)
A VOT (VOR Test facility) radiates a **continuous Morse dots tone**. Used on the ground to verify aircraft VOR receiver accuracy. Aircraft must read 000°(±4°) FROM or 180°(±4°) TO.

Q6. Maximum permissible bearing error when checking VOR airborne equipment using a VOT:

(a) ±4°

(b) ±3°

(c) ±5°

(d) ±2°

Answer: (a)
VOT maximum permissible error = **±4°**. Exceeding this requires equipment servicing before use.

Q7. The DOC of a VOR is valid:

(a) During daylight hours only

(b) During night hours only

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(c) During twilight periods only

(d) Both day and night

> **Answer: (d)**
> VOR DOC is valid **day AND night**. VHF is immune to skywave propagation. Contrast with NDB/ADF whose DOC is day only.

Q8. Total system accuracy of a VOR is:

(a) ±1°

(b) ±3°

(c) ±5°

(d) ±10°

> **Answer: (c)**
> Total VOR accuracy = **±5°**: site error ±1° + propagation/scalloping ±1° + airborne equipment ±3° (RSS combination).

Q9. Flying directly over a VOR station, the indications are:

(a) OFF flag appears, TO/FROM alternates, CDI oscillates (cone of confusion)

(b) TO flag, CDI centres

(c) Bearing becomes more accurate

(d) No change — VOR is accurate overhead

> **Answer: (a)**
> Overhead VOR = **cone of confusion**: OFF flag may appear, TO/FROM alternates rapidly, CDI needle oscillates. Normal and temporary.

Q10. On a standard CDI, angular value of one dot deflection:

(a) 5°

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(b) 1°

(c) 2°

(d) 10°

> **Answer: (c)**
> CDI sensitivity = **2° per dot**. Full scale = 5 dots × 2° = ±10°.

Q11. Full Scale Deflection on a standard CDI:

(a) ±5°

(b) ±10°

(c) ±20°

(d) ±2°

> **Answer: (b)**
> CDI FSD = **±10°** (5 dots × 2°/dot).

Q12. Which RMI statement is correct?

(a) Arrowhead points into the tailwind

(b) Tail indicates QDM

(c) Arrowhead indicates QDM (magnetic bearing TO station)

(d) RMI shows track made good

> **Answer: (c)**
> RMI: arrowhead (head) = **QDM** = bearing TO station. Tail = QDR = radial = bearing FROM station.

Q13. Compared to CVOR, a DVOR:

(a) Operates at higher frequencies

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(b) Uses a single rotating antenna

(c) Has a larger cone of confusion

(d) Has less site error due to its electronically switched antenna array

> **Answer: (d)**
> DVOR uses a ring of 48 antennas switched electronically. Doppler effect generates FM variable signal. **Less susceptible to site errors** from terrain/structures.

Q14. Aircraft on 090° radial from VOR. Station variation = 6°E. The QTE is:

(a) 084°T

(b) 096°T

(c) 090°T

(d) 264°T

> **Answer: (b)**
> QDR = 090°M. Apply variation at VOR: 090° + 6° (East) = **096°T**. (True = Magnetic + East variation)

Q15. Max beacon spacing for airway with ±5 NM half-width, VOR error 5°:

(a) 120 NM

(b) 60 NM

(c) 150 NM

(d) 100 NM

> **Answer: (a)**
> Max spacing = (half-width × 60) / error = (5 × 60) / 5 × 2 = 600/5 = **120 NM**. (Using the standard 600/error_degrees formula.)

Q16. Max beacon spacing if VOR bearing error is 5.5°:

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(a) 120 NM

(b) 100 NM

(c) 109 NM

(d) 80 NM

> **Answer: (c)**
> Max spacing = 600 / 5.5 ≈ **109 NM**.

Q17. Number of VOR channels in the 108–112 MHz band:

(a) 80 total

(b) 160 total

(c) 40 VOR channels (even tenths only)

(d) 20 VOR channels

> **Answer: (c)**
> 108–112 MHz: even tenths = VOR (**40 channels**); odd tenths = ILS Localiser (40 channels).

Q18. Total number of VOR channels:

(a) 200

(b) 160

(c) 120

(d) 40

> **Answer: (b)**
> Total VOR channels = 40 (shared, 108–112) + 120 (VOR only, 112–118) = **160**.

Q19. Aircraft heading 287°, starboard drift 14°. Track made good (radial being flown):

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(a) 273°

(b) 287°

(c) 301°

(d) 315°

> **Answer: (c)**
> Track = Heading + Starboard drift = 287° + 14° = **301°**. QDR = track when flying outbound from VOR.

Q20. Aircraft heading 287°, 14° starboard drift. VOR phase difference reading:

(a) 273°

(b) 287°

(c) 301°

(d) 315°

> **Answer: (c)**
> Phase difference = radial = QDR = Track = 287° + 14° = **301°**.

Q21. CDI shows TO indication when:

(a) Aircraft is heading toward the station

(b) Aircraft is closer than 10 NM

(c) QDM is within ±80° of OBS-selected course

(d) Aircraft is below the cone of confusion

> **Answer: (c)**
> TO flag illuminates when **QDM (bearing to station) is within ±80° of the OBS setting**. Not dependent on aircraft heading — geometry only.

Q22. Aircraft on 152° radial. OBS set to 329° (TO). CDI indication:

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(a) Bar deflects RIGHT — approximately 1.5 dots

(b) Bar deflects LEFT — approximately 1.5 dots

(c) Bar centred

(d) OFF flag

> **Answer: (a)**
> OBS 329° TO means desired radial = 149°. Aircraft is on 152° — 3° clockwise of 149° → aircraft is LEFT of the 329° inbound course → bar deflects **RIGHT** (fly right). 3° ÷ 2°/dot = 1.5 dots.

Q23. At 100 NM from VOR with ±5° total error, lateral displacement is approximately:

(a) 5 NM

(b) 7.5 NM

(c) 8.3 NM

(d) 8.7 NM

> **Answer: (d)**
> Exact: 100 × sin(5°) = 100 × 0.0872 ≈ **8.7 NM**. Using 1:60: 100 × 5/60 ≈ 8.3 NM.

Q24. VOR variation should be applied at:

(a) Aircraft position

(b) Mid-point between aircraft and station

(c) The VOR station

(d) The destination aerodrome

> **Answer: (c)**
> For VOR, variation is applied at the **VOR station**. Contrast with NDB/ADF: variation applied at the aircraft.

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Q25. Aircraft on 045° radial. RMI shows:

(a) Head at 045°, tail at 225°

(b) Head at 225°, tail at 045°

(c) Head and tail both at 045°

(d) Head at 225°, tail at 225°

> **Answer: (b)**
> QDR (radial) = 045° → RMI tail = 045°. RMI head = QDM = 225° (reciprocal). Answer: head 225°, tail 045°.

Q26. Standard HSI CDI is set to 5°/dot. Full scale deflection is:

(a) 5°

(b) 10°

(c) 20°

(d) 15°

> **Answer: (b)**
> Standard HSI: 5°/dot × 2 dots = **10°** FSD.

Q27. VOR ident transmission interval:

(a) Every 30 seconds

(b) Every 5 seconds

(c) Every 10 seconds

(d) Every 15 seconds

> **Answer: (c)**
> VOR Morse ident is transmitted every **10 seconds**.

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Q28. If the VOR auto-monitor detects bearing error > ±1°:

(a) Station automatically shuts down or switches to standby, suppressing ident

(b) Transmits a warning tone

(c) Continues at reduced power

(d) Changes frequency

> **Answer: (a)**
> Auto-monitor: if error > ±1° → **station shuts down or switches to standby**. Ident suppressed for 30 s. Never use VOR without ident.

Q29. VORTAC provides:

(a) VOR bearing for civil aircraft and TACAN bearing + DME for military aircraft

(b) VOR bearing only

(c) DME for civil and military only

(d) ILS and VOR combined

> **Answer: (a)**
> VORTAC: civil aircraft receive VOR azimuth + DME distance (via TACAN DME on paired freq). Military: TACAN bearing + DME.

Q30. VOR emission designator:

(a) A9W

(b) A2A

(c) N0NA1A

(d) P0N

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Answer: (a)
VOR = **A9W**. A = AM DSB; 9 = composite signal; W = combination of telephony, telegraphy and navigation.

Q31. HSI (5°/dot), aircraft is 4° to the right of selected course. CDI bar:

(a) Right by 2 dots

(b) Left by 2 dots

(c) Left, less than 1 dot

(d) Right, less than 1 dot

Answer: (c)
5°/dot HSI. Aircraft 4° RIGHT of course → bar deflects LEFT (fly left to intercept). Magnitude = 4° ÷ 5°/dot = 0.8 dots left = less than 1 dot LEFT.

Q32. DVOR differs from CVOR in that:

(a) Different frequencies

(b) Uses Doppler effect with an array of antennas to generate FM variable signal; AM is the reference

(c) Larger DOC

(d) Uses pulse modulation

Answer: (b)
DVOR: AM carrier = reference; electronically switched 48-antenna ring generates FM variable via Doppler effect. CVOR: FM subcarrier = reference; AM rotating limàcon = variable.

Q33. Aircraft on radial 270°, heading 090° inbound to VOR, OBS set to 090°. TO/FROM indicator shows:

(a) TO

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(b) FROM

(c) OFF

(d) Alternating TO/FROM

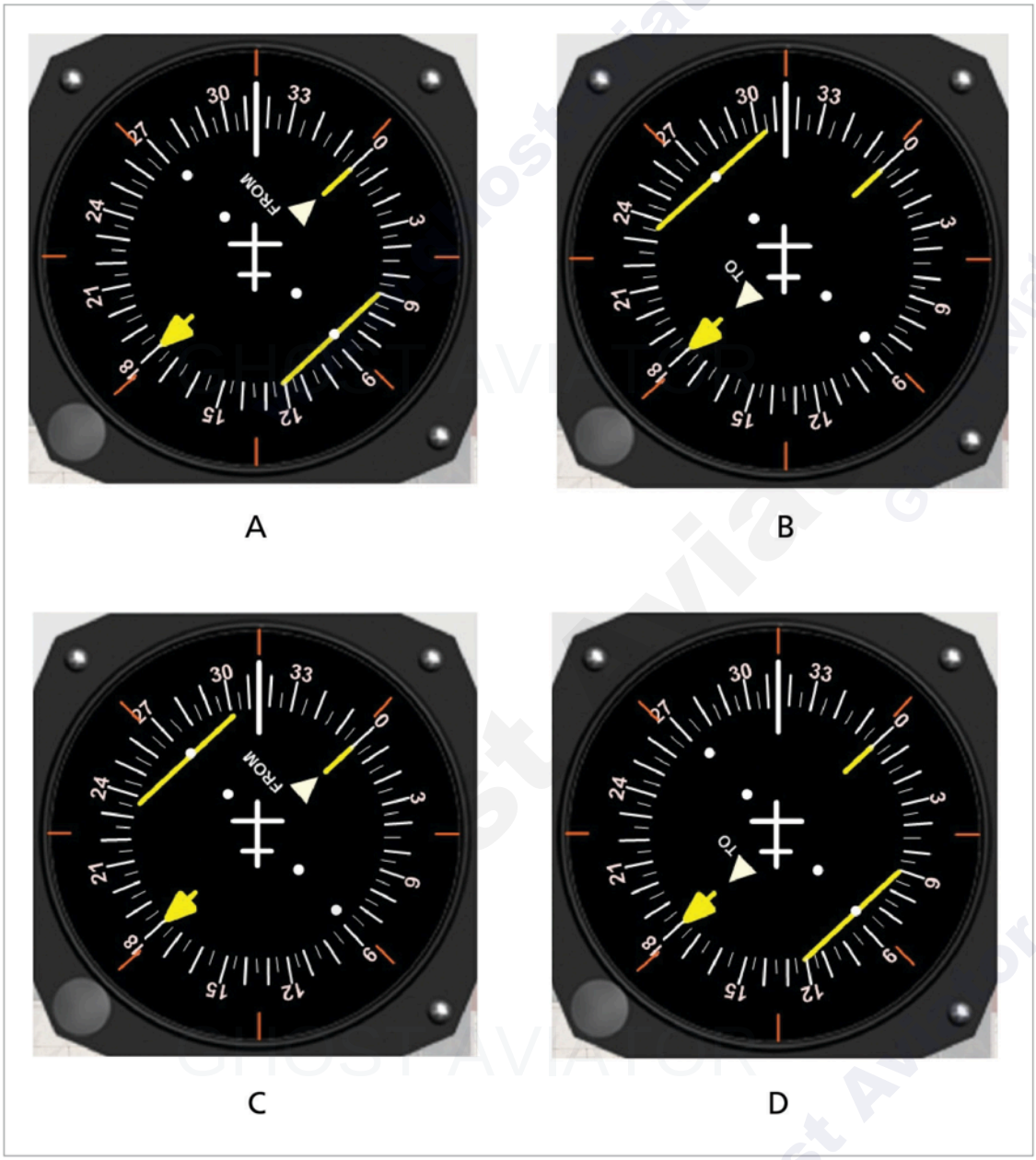
> **Answer: (a)**
> Aircraft west of VOR flying east (inbound). QDM = 090°. OBS = 090°. QDM is within 80° of OBS → **TO**.

Annex A — HSI Diagrams (Q31)

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b: B
c: C
d: D

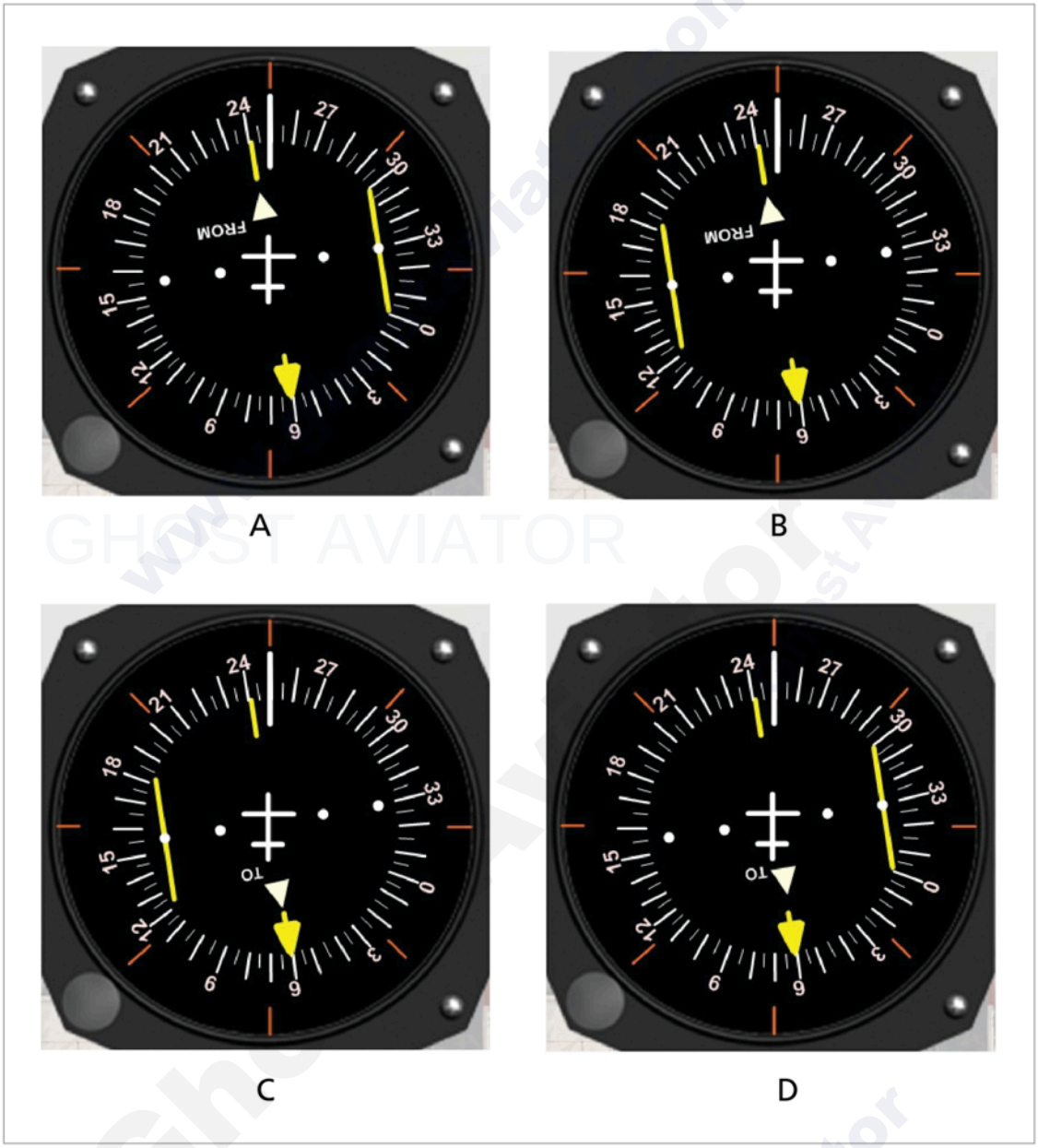
Annex A



Annex A: HSI options for Question 31

Annex B — HSI Diagrams (Q32)

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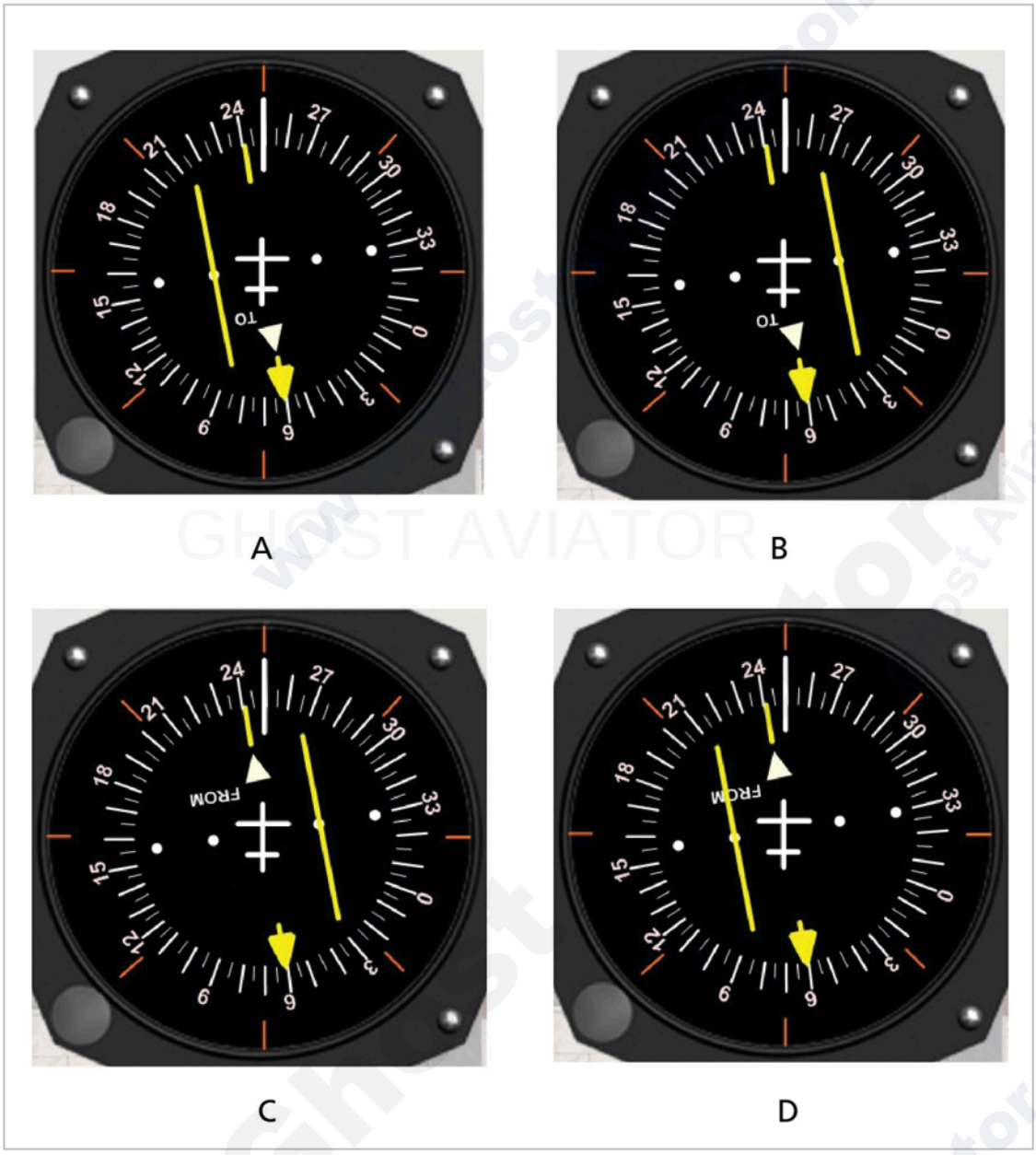


Annex B: HSI options for Question 32

Annex C — HSI Diagrams (Q33)

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Annex C



Annex C: HSI options for Question 33

Additional Question Diagrams

Q3 — RMI Options

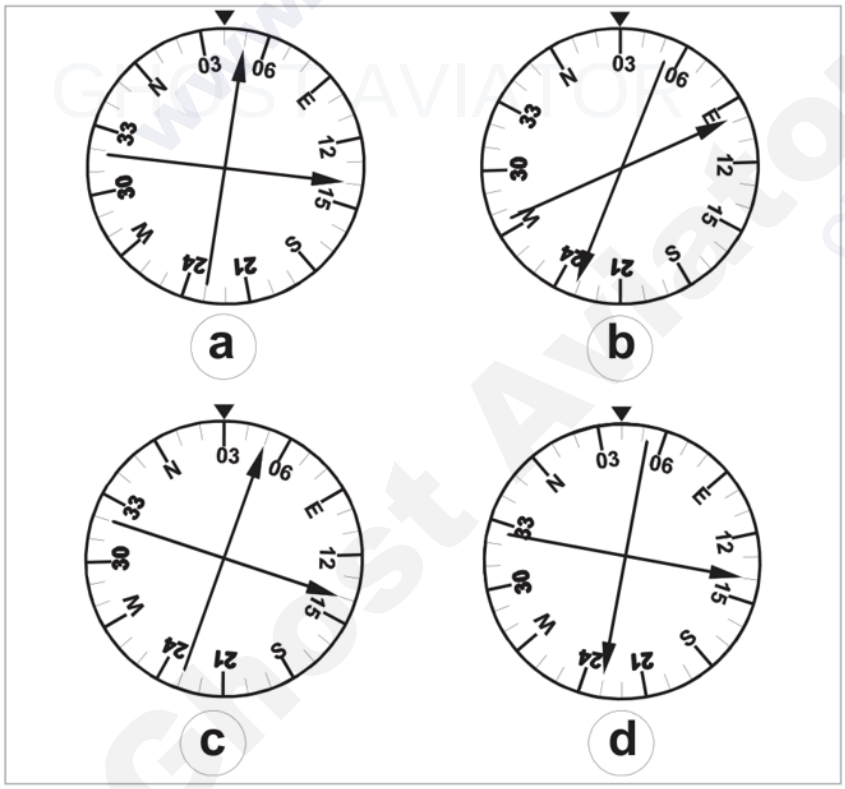
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a. 54.5 NM
b. 109 NM
c. 66 NM
d. 132 NM

2. The Designated Operational Coverage quoted for VOR beacons in the COMM section of the AIP:

a. is only applicable by day
b. guarantees a protection ratio of at least 3 to 1 by day and night
c. defines the airspace within which an aircraft is assured of protection from interference from other VORs on the same channel
d. is determined by the type of surface over which the signal will have to travel

3. An aircraft is tracking away from a VOR on the 050 radial with 10° starboard drift. An NDB lies to the east of the VOR. Which of the RMIs illustrated below shows the aircraft when it is obtaining a relative bearing of 100° from the NDB?

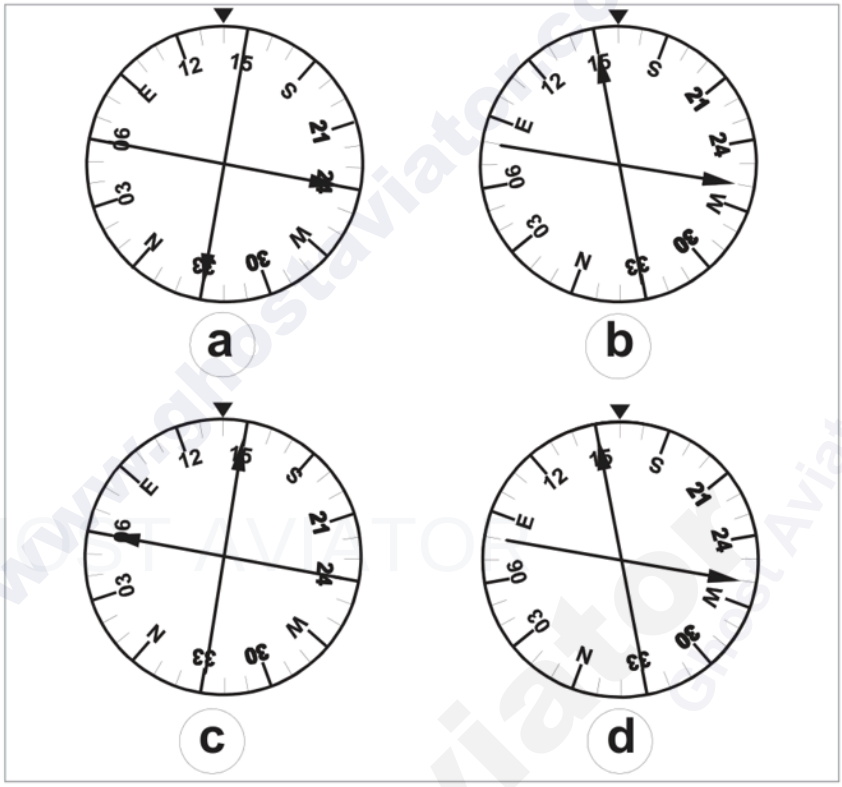


Q3: Four-option RMI diagram

Q15 — RMI Options

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NDB lies to the south of the VOR. Which of the RMIs illustrated below shows the aircraft when it is obtaining a relative bearing of 100° from the NDB?



16. **Assuming the maximum likely error in VOR to be 5°, what is the maximum distance apart that beacons can be situated on the centre line of a UK airway in order that an aircraft can guarantee remaining within the airway boundary?**

a. 60 NM
b. 100 NM
c. 120 NM
d. 150 NM

17. **AN aircraft, heading 150°, is 100 NM north of a VOR, the pilot intends to home to the VOR on the 030 radial. The pilot should set ..... on the OBS and on reaching the 030 radial should turn ..... onto a heading of ....., assuming zero wind.**

a. 210 left 030
b. 030 right 210
c. 210 right 210
d. 150 left 210

18. **The type of emission radiated by a VOR beacon is:**

a. a double channel VHF carrier with one channel being amplitude modulated and the second channel being frequency modulated
b. a single channel VHF carrier wave amplitude modulated at 30 Hz with a sub carrier being frequency modulated at 30 Hz
c. a VHF carrier wave with a 90 Hz frequency modulation and a 150 Hz amplitude modulation

Q15: Four-option RMI diagram

Q25 (RMI) and Q26 (CDI)

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answer the following: The information given on the RMI indicates:

a. that the aircraft is heading 033°(M), is on the 310° radial from the VOR, and bears 050°(M) from the NDB
b. that the aircraft is heading 330°(M), is on the 310° radial from the VOR, and bears 050° from the NDB
c. that the aircraft is heading 330°(M), is on the 130° radial from the VOR, and bears 050°(M) from the NDB
d. that the aircraft is heading 330°(M), is on the 130° radial from the VOR, and bears 230°(M) from the NDB



26. The VOR in an aircraft is correctly tuned and set to define the centre line of an airway within UK airspace which you intend to fly. The indication received on the VOR/ILS deviation indicator is shown to the right. At the same time the DME gave a range of 90 NM from the facility. At the time of the observation, the aircraft's radial and distance from the airway centre line were:

a. 062 radial 9 NM
b. 074 radial 6 NM
c. 242 radial 6 NM
d. 254 radial 9 NM



27. The normal maximum error which might be expected with a VOR bearing obtained within the DOC is:

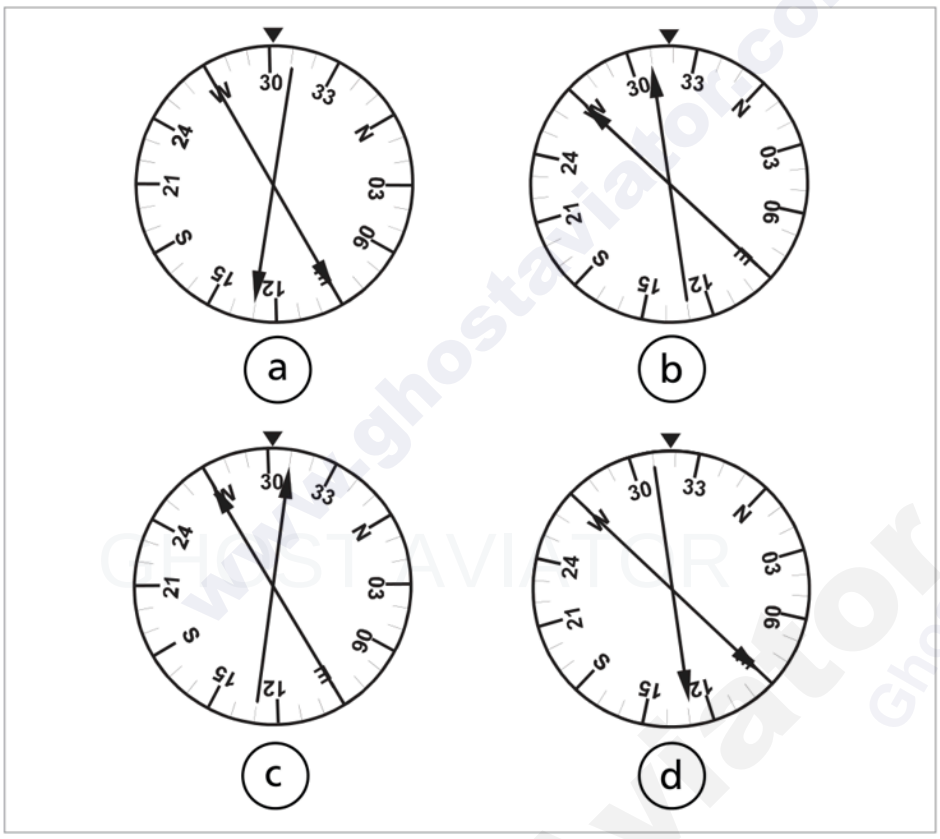
a. plus or minus 1°
b. plus or minus 2°
c. plus or minus 5°
d. plus or minus 10°

Q25 RMI and Q26 CDI diagrams

Q28 — RMI Options

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is on its present track with a QDR from "X" of 270°?



29. **The VOR indications on an RMI whose deviation is not zero:**

a. are magnetic
b. are compass
c. are relative
d. must have deviation applied before being used

Q28: Four-option RMI diagram

Q30 — CDI Options

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(assume 1 dot = 2°)



31. **Using Annex A.**
An aircraft is flying on the 170 radial with a heading of 315°(M).
The course on the HSI is set to 180°.
Which HSI shows the correct indications?

a: A
b: B
c: C
d: D

32. **Using Annex B.**
An aircraft is flying on the 050 radial with a heading of 250°(M).
The course on the HSI is set to 060°.
Which HSI shows the correct indications?

a: A
b: B
c: C
d: D

Q30: Four-option CDI diagram

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